Literature Status: Tingley's Problem for Low-Rank Cartan Factors

Run `fa_banach_001` — lane 9

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Status: literature already answered. Problem 3.20 of Peralta–Tanaka (arXiv:1608.06327) asks whether a surjective sphere isometry between Cartan factors of rank at most four extends to a surjective real linear isometry. Fernández-Polo–Peralta (arXiv:1611.10218, Theorem 2.5) explicitly complete the cases left open there and prove the stronger theorem for arbitrary weakly compact JB*-triples.

1 Original question

For a Banach space X, write $S(X)$ for its unit sphere. After proving extension results for compact C*-algebras and for higher-rank JB*-triples, Peralta and Tanaka write "We do not know the answer in the remaining cases" and pose the following on printed page 17 of *A solution to Tingley's problem for isometries between the unit spheres of compact C*-algebras and JB*-triples* [1].

Source Problem 3.20. *Let C and B be Cartan factors of rank at most 4, and suppose that $f : S(C) \to S(B)$ is a surjective isometry. Does f extend to a surjective real linear isometry from C into B?*

This is a sharply delimited low-rank operator-space instance of Tingley's problem, not the unrestricted Banach-space problem.

2 Separate later answer

Fernández-Polo and Peralta explicitly cite the Peralta–Tanaka paper and state that ranks 2 through 4 were left open there. Their introduction then says that they complete that study by solving all remaining low-rank cases [2]. Their Theorem 2.5, on printed page 8, is stronger than the source question.

Answering Theorem 2.5. *Let $f : S(E) \to S(B)$ be a surjective isometry between the unit spheres of two weakly compact JB*-triples. Then there exists a surjective real linear isometry $T : E \to B$ satisfying $T|_{S(E)} = f$.*

Cartan factors are covered by the paper's elementary JB*-triple analysis. Thus applying Theorem 2.5 to the two Cartan factors gives exactly the extension asked for in Problem 3.20. The paper also proves the more structured elementary-factor conclusion in Theorem 4.1: at rank at least two, the extension can be chosen complex linear or conjugate linear. Theorem 4.7 directly treats finite-dimensional elementary JB*-triples.

The logical match is therefore exact and author-explicit: the later paper identifies the earlier open cases, announces that it completes them, and proves a theorem with strictly broader hypotheses.

3 Scope and search evidence

A bounded search through 11 August 2026 checked the run registry and solution indexes, the exact source wording, citations of arXiv:1608.06327, and later work on low-rank Cartan factors and Tingley's problem. The decisive match is the separate paper arXiv:1611.10218 and its published version. Its abstract calls the result a complete solution for weakly compact JB*-triples, its introduction explicitly identifies the cases left open in the source, and Theorem 2.5 supplies the extension.

This packet classifies only Problem 3.20. It does not claim a solution of Tingley's problem for arbitrary Banach spaces.

References

[1] A. M. Peralta and R. Tanaka, *A solution to Tingley's problem for isometries between the unit spheres of compact C*-algebras and JB*-triples*, Sci. China Math. **62** (2019), 553–568; arXiv:1608.06327.

[2] F. J. Fernández-Polo and A. M. Peralta, *Low rank compact operators and Tingley's problem*, Adv. Math. **338** (2018), 1–40; arXiv:1611.10218, doi:10.1016/j.aim.2018.08.018.